

What Kills? Factors that Affect State Death Rate

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Introduction

- National Health Expenditure (NHE) is one of the most important economic metrics in the US and it is used to allocate funds to different sectors to improve the health of the country (CMS, 2021).
- In this project we are working to find what variables most heavily affect death rate in the United States to inform funding.
- NHE is funded by federal taxes of which 29% is going to health programs and services spending (KFF, 2021).
- This is an interesting topic to research especially after COVID to see the updated weight of factors that lead to death rate.
- Purpose:** Give insight into important problems and solutions that the United States Public Health Service should prioritize.

Research Questions

- Does higher Medicaid spending per capita lead to lower death rates?
- Did lifting COVID restrictions as of May 2021 (“Yes”) result in higher death rates?
- Does lower Medicare enrollment in states lead to higher death rates?

Data Summary

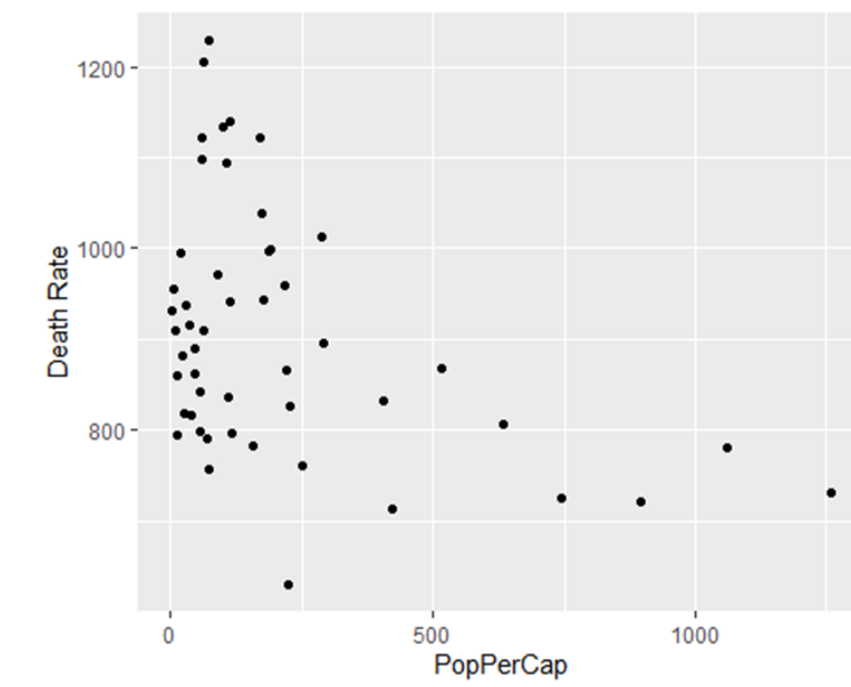
- Our data was collected from the following sources:
 - KFF (Kaiser Family Foundation)
 - The CDC
 - CMS (The Center for Medicare and Medicaid Services)
 - The Cook Political Report
 - New York Times
- The data included 50 observations (states)
- Each variable came from a different source and was cleaned and merged into the larger data set

Variable Name	Abbreviated Name	Description	Type
Death Rate	DeathRate	Death rate per 100,000 residents by state	Quantitative [Response]
Drug Overdose Death Rate	OverdoseDeaths	Drug overdose deaths per 100,000 residents by state	Quantitative
Obesity Prevalence	ObesityPrev	Prevalence of obesity based on self-reported weight and height by state	Quantitative
Total Medicare Enrollment	MedicareEnr	Medicare enrollment as a percent of the resident population by state	Quantitative
Total Medicaid Spending per Capita	MedicaidSpend	Total Medicaid expenditure per capita (total divided by population) by state	Quantitative
Lifted COVID restrictions	LiftCOVID	If “yes”, then in general-mask mandates, stay-at-home orders or interstate-quarantines in-place, most businesses open by state. If “no”, restrictions still in place	Qualitative
State Population density	PopPerCap	Population density in the U.S. by states	Quantitative
Senior Population Level	SeniorPop	Percentage of the state population with individuals 65 years or older split into three quantiles and labeled “Low”, “Medium”, and “High”.	Quantitative
Senior Population Level (Interval Form)	Senior.Pop1	Senior Population Level shown in continuous interval form [11.6,16.6], (16.6,17.9], (17.9,21.7] *Not used in analysis but included to support interpretation of SeniorPop	Qualitative
Political Majority	PoliticalMaj	Assigns either R (for Republican) or D (for Democrat) based on voter data from the 2020 presidential election	Qualitative

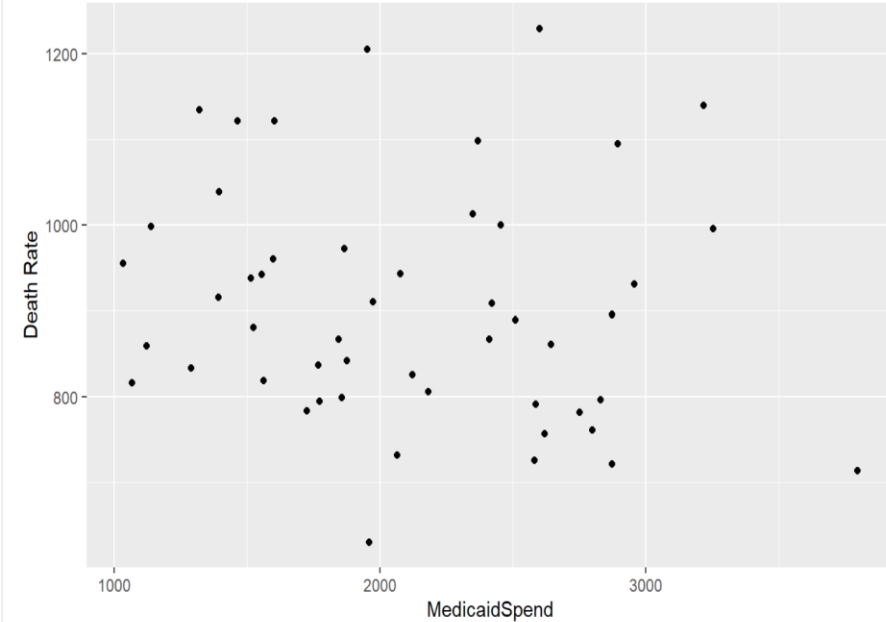
Exploratory Data Analysis

Correlation between Obesity Prevalence and Death Rate: 0.783

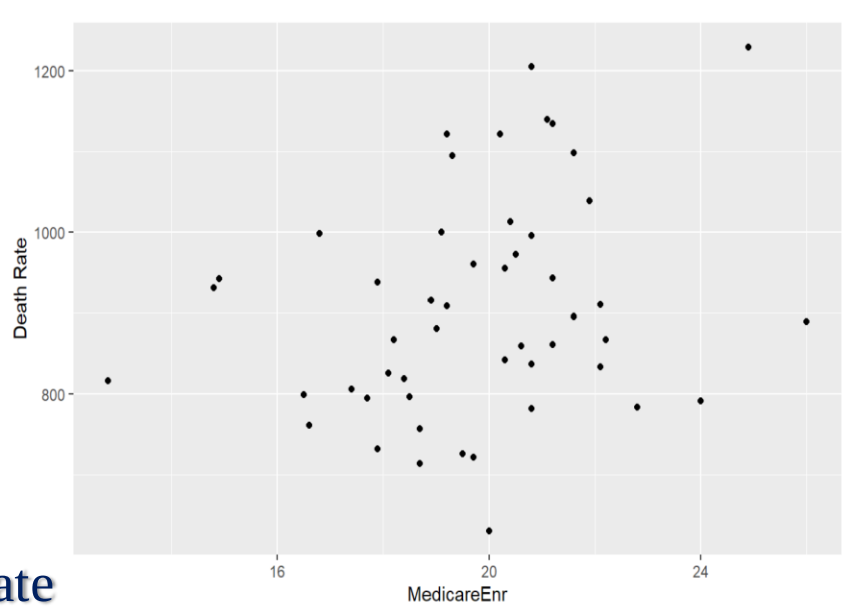
High Order Term PopPerCap



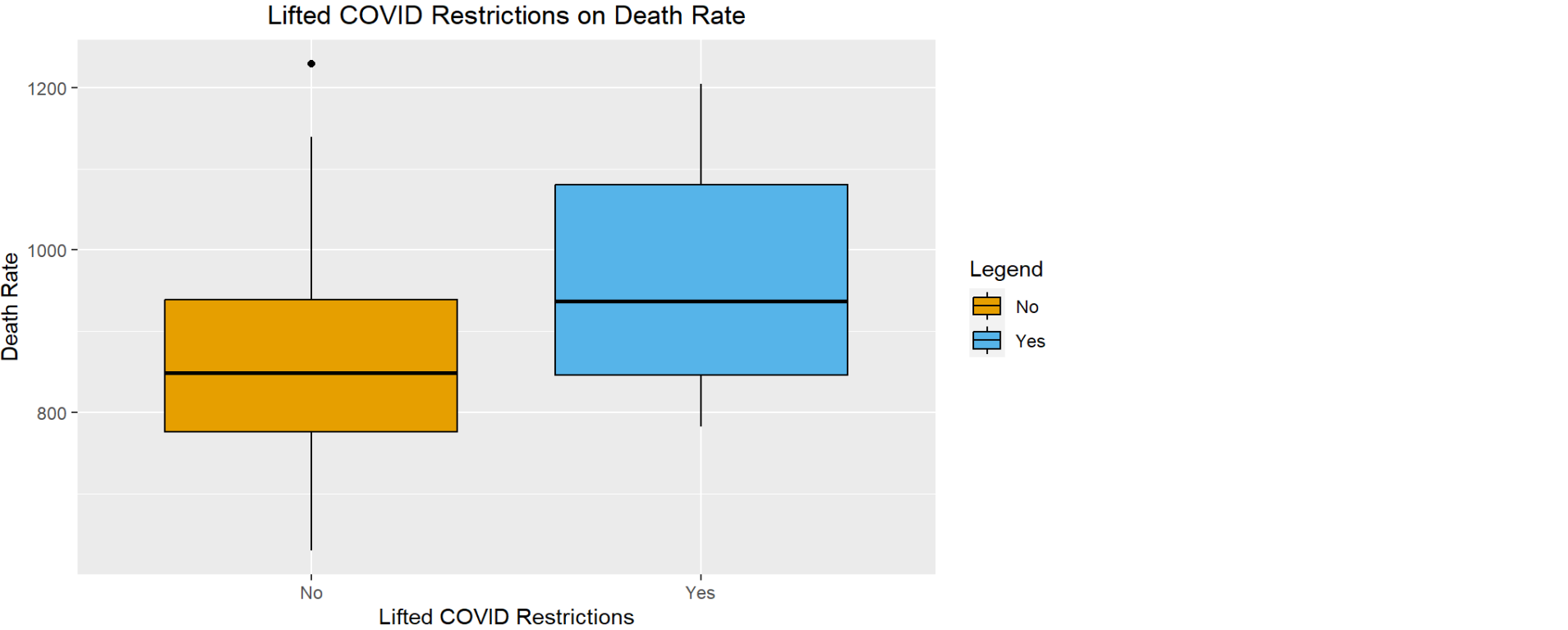
Medicaid Spending on Death Rate



Medicare Enrollment on Death Rate

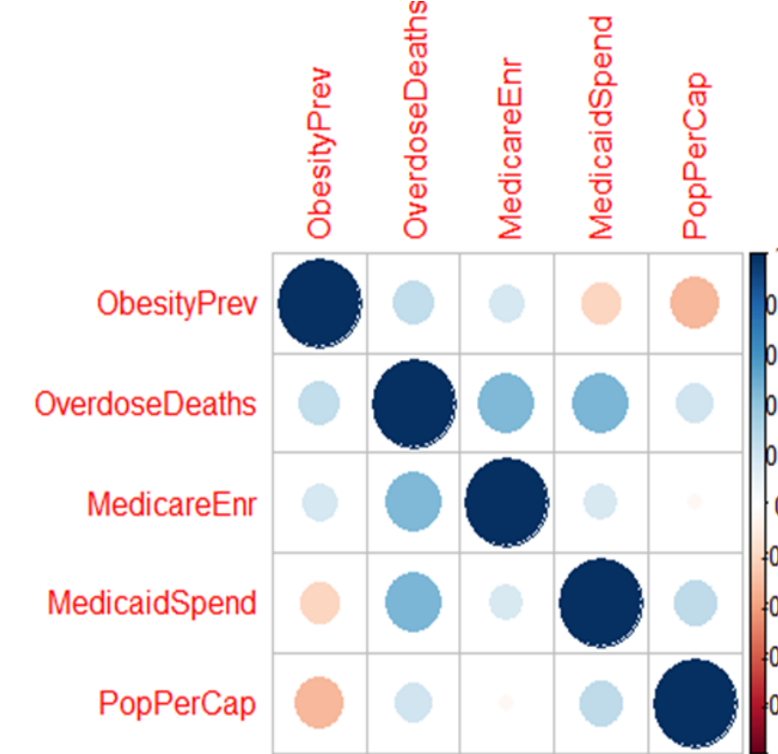


Lifted COVID Restrictions on Death Rate



Multicollinearity & Variable Screening

Multicollinearity:



- No individual VIFs greater than 10
- Average VIF less than 3
- Conclusion:** No evidence of severe multicollinearity

Stepwise Regression: prem=0.3; pent=0.1

Stepwise Selection Summary					
Step	Variable	Added/Removed	R-Square	Adj. R-Square	C(p)
1	ObesityPrev	addition	0.613	0.605	17.3370
591.4924	86.1764				
2	OverdoseDeaths	addition	0.666	0.652	10.6490
586.1155	80.8952				
3	PopPerCap	addition	0.725	0.707	3.0050
578.4083	74.2053				

Weighted Least Squares

Weight Formula: $w_i = \frac{1}{(\hat{\epsilon}_i)^2}$

- w_i represents the weight for the i -th observation.
- $\hat{\epsilon}_i$ represents the fitted residual for the i -th observation.

*Note: Statewide data warranted the use of weighted least squares regression as some observations (states) may have larger associated values due to size.

Model Building

Stage One: Quantitative Variables

Initial

$$\text{DeathRate} = \beta_0 + \beta_1 \text{ObesityPrev} + \beta_2 \text{OverdoseDeaths} + \beta_3 \text{MedicareEnr} + \beta_4 \text{MedicaidSpend} + \beta_5 \text{PopPerCap}$$

Final

$$\text{DeathRate} = \beta_0 + \beta_1 \text{ObesityPrev} + \beta_2 \text{OverdoseDeaths} + \beta_3 \text{PopPerCap} + \beta_4 \text{PopPerCap}^2$$

Stage Two: Qualitative Variables

Initial

$$\text{DeathRate} = \beta_0 + \beta_1 \text{ObesityPrev} + \beta_2 \text{OverdoseDeaths} + \beta_3 \text{PopPerCap} + \beta_4 \text{PopPerCap}^2 + \beta_5 \text{SeniorPopM} + \beta_6 \text{SeniorPopH} + \beta_7 \text{PoliticalMajR} + \beta_8 \text{LiftCOVIDY}$$

*Where SeniorPopM = {1 if Medium, 0 otherwise}; SeniorPopH = {1 if High, 0 otherwise}; PoliticalMajR = {1 if Republican, 0 otherwise}; LiftCOVIDY = {1 if Yes, 0 otherwise}

Final

$$\text{DeathRate} = \beta_0 + \beta_1 \text{ObesityPrev} + \beta_2 \text{OverdoseDeaths} + \beta_3 \text{PopPerCap} + \beta_4 \text{PopPerCap}^2 + \beta_5 \text{LiftCOVIDY}$$

Stage Three: Interactions

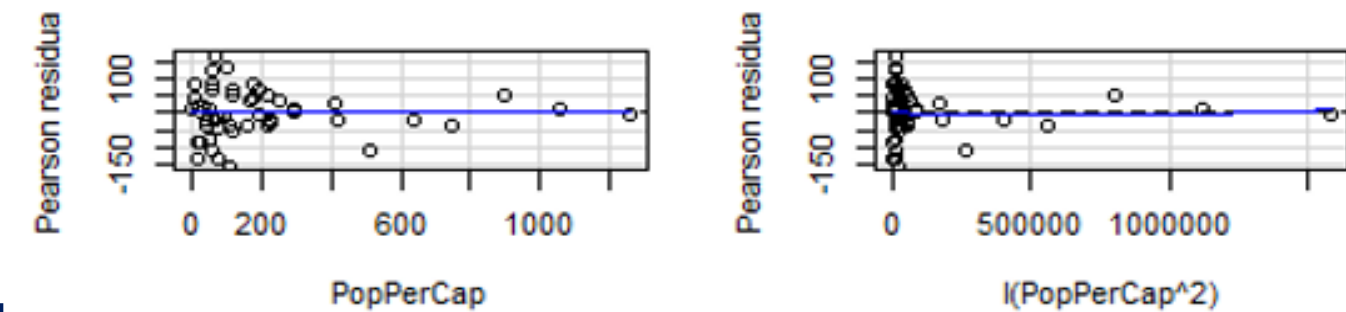
No notable interactions were observed or are of interest as evidenced by the interaction EDA.

Assumptions:

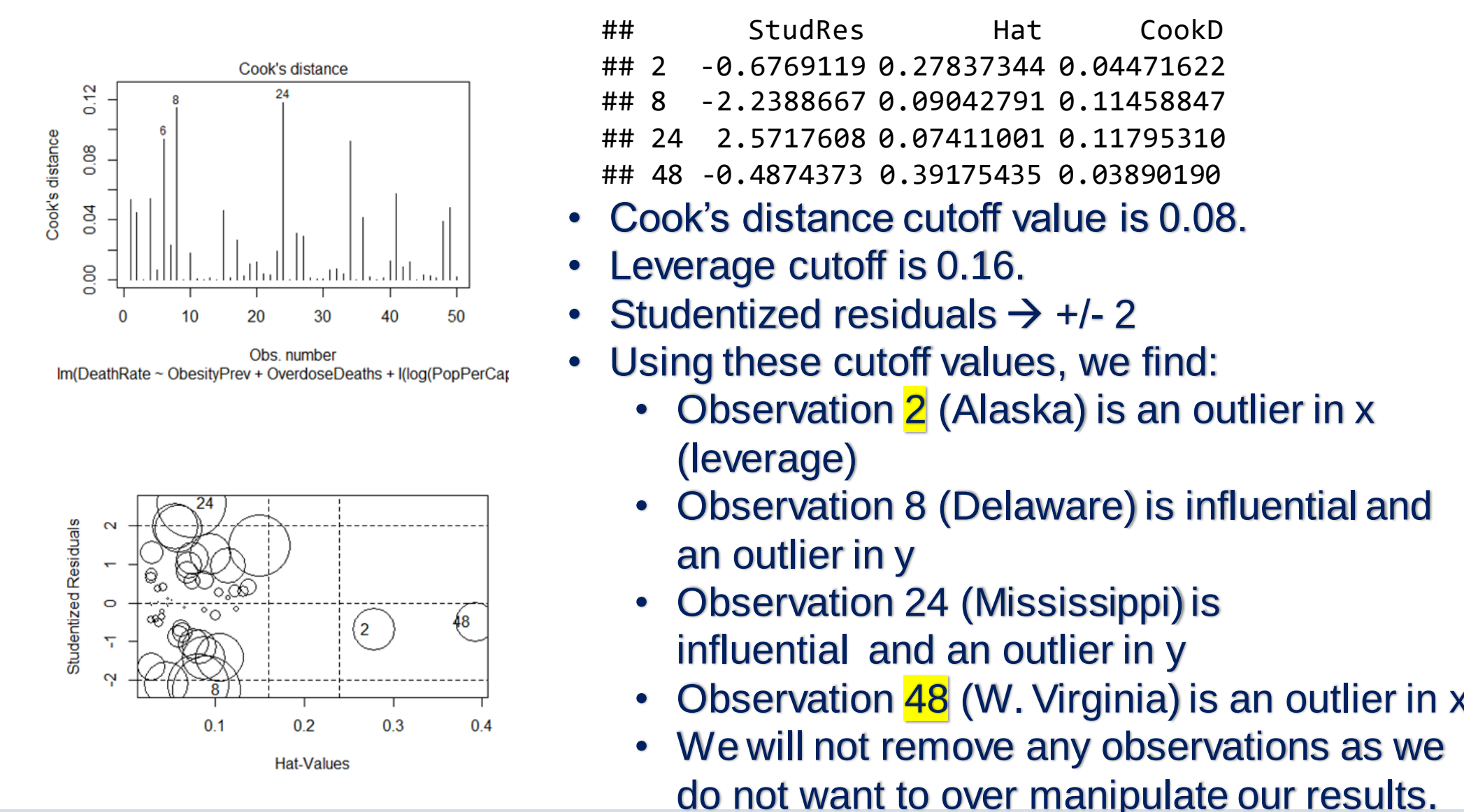
- Fit:** MET
- Constant Variance:** VIOLATED
- Normality:** MET
- Independence:** Not applicable (No time-series data)

Transformation:

$$\beta_4 \text{PopPerCap}^2 \rightarrow \beta_4 \log(\text{PopPerCap})$$



Influential Observations



- Observation 2 (Alaska) is an outlier in x (leverage)
- Observation 8 (Delaware) is influential and an outlier in y
- Observation 24 (Mississippi) is influential and an outlier in y
- Observation 48 (W. Virginia) is an outlier in x
- We will not remove any observations as we do not want to over manipulate our results.

FINAL MODEL

$$\text{DeathRate} = \beta_0 + \beta_1 \text{ObesityPrev} + \beta_2 \text{OverdoseDeaths} + \beta_3 \log(\text{PopPerCap}) + \beta_4 \text{LiftCOVIDY}$$

Conclusion

Prediction Equation:

$$\hat{Hat}\{\text{Deathrate}\} = 227.65 + 18.74 \text{ObesityPrev} + 4.20 \text{OverdoseDeaths} - 26.40 \log(\text{PopPerCap}) + 43.91 \text{LiftCOVID}$$

Interpretation: For every one unit increase in obesity prevalence, the death rate per 100,000 residents by state increases by 18.74. For every one unit increase in drug overdose deaths per 100,000 residents by state, the death rate increases by 4.20. For every one unit in state population density, death rate decreases by log of 26.40. The death rate for states with COVID restrictions lifted is 43.91 units higher than states that had not lifted restrictions as of May 3rd 2021. This model is useful when trying to predict a death rate within a certain state.

Efficacy: This model accounts for 74.83% of the variation in the data with a Root MSE value of 1.282 and a p-value of <0.00001.

Limitations: The log transformation of the explanatory variable PopPerCap reduces the interpretability of our model.

Usage Example: For our example, we wanted to test the accuracy of our equation by predicting the death rate in Virginia. Our equation predicts the death rate in Virginia each year is about 954 people per 100,000 and the actual death rate in Virginia is about 867 people per 100,000. This gives a residual value of 87 and error % of 10%.

Improvements & Future Research

- Select different variables and find more variables that prove to be significant and contribute towards the model.
- Try to find more qualitative predictors to not have to manipulate senior population into a qualitative predictor and instead leave it as a quantitative predictor.
- Replicate this research with 2022 data to gage what death rate looks like more recently.

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